9.3.3 Classification of Mycetoma

The Mycetoma is classified on the basis of the causative agent **aerobic actinomycetes** causes **actinomycetoma** whereas **hyaline and phaeoid fungi** cause **eumycetoma.**

9.3.4 Clinical Features

The clinical entity depends upon the age of the lesions and to size, shape and color of the grains. The painless localized swollen lesions with purulent fluid lead to the secondary bacterial infections. Important features of Mycetoma are as follows:

i. Tumor like swelling

ii. Multiple draining sinuses

iii. Presence of grains or granules in sinuses.

9.3.5 Laboratory Diagnosis

i. Samples

The clinical sample in Mycetoma is usually grains, pus exudates or biopsy were collected.

a. Direct Examination

Grams staining, modified Ziehl – Neelson staining, LPCB and KOH wet mount are used to visualize the organisms.

The grains should be washed, crushed and cultured on different media. Crushed grains are examined (Figure 9.8a).

KOH mount

Eumycotic grains show thick 2–6 µm hyphae with large globose swollen cells with or without chlamydospores. Actinomycotic grains show thin filaments of 0.5–1 µm with coccoid or bacillary forms.

Gram stain

Actinomycetoma grains show Gram-positive branching filamentous bacteria with branches (Figure 9.8b).

Ziehl - Neelson stain

Nocardia species show red pink acid fast filamentous bacteria.

b. Culture

Crushed grains are washed several times with normal saline without antibiotics and inoculated on to **Sabouraud dextrose agar, blood agar, Lowenstein -Jonson** media and brain-heart infusion agar. The plates are incubated at 25°C, 37°C and 44°C for various organisms (Figure 9.8c).

ii. Treatment

1. Ketoconazole 200 mg and Itraconazole 100mg are given for 8–24 months to treat eumycetoma.
2. Sulfonamides, tetracylines, streptomycin, amoxicillin are administered to treat actinomycetoma.



(a) Biopsy-Black grains



(b) Microscopic Morphology



(c) Macroscopic Morphology

Figure 9.8: Laboratory Diagnosis of Mycetoma

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